

Data. Hazardous prompts were taken from WMDP-Bio question stems (1,273 examples), excluding answer choices. For safe examples I used MMLU, including both general subjects and biological subjects (anatomy, college biology, medical genetics), an initial version excluded biology entirely, which caused models to learn "biology = refuse" rather than the meaningful distinction. I later added PubMedQA (~1,000 biomedical research questions) to address a class imbalance toward the hazardous class and reduce over-refusal on safe biology prompts. Natural Questions and SQuAD were considered but excluded due to disk space constraints.

Modelling. I started with a TF-IDF + Logistic Regression baseline, then fine-tuned four transformers: DistilBERT (as suggested in the task spec), BERT, RoBERTa, and BioBERT. My hypothesis was that RoBERTa and BioBERT would outperform due to larger pretraining corpora and domain specificity respectively. DeBERTa-v3-small was attempted but produced NaN loss under both fp16 and bf16 on Python 3.13 and was excluded. All transformers were trained for 3 epochs, learning rate 2×10^{-5} (1×10^{-5} for RoBERTa), batch size 16, max sequence length 256.

Results. All transformers exceeded F1 0.99 versus 0.8951 for logistic regression. Logistic regression produced ~16 false negatives versus 1-2 for transformers, a critical gap in a safety context. On a manually constructed adversarial set of 14 prompts (7 rephrased hazards, 7 safe biology questions), all transformers caught 100% of rephrased hazards. The universal failure mode was over-refusal on safe biology vocabulary like *"How does the immune system fight infection?"* was refused by every model in every configuration. Counterintuitively, BioBERT performed worst on safe biology questions before PubMedQA, suggesting its biomedical pretraining caused it to associate academic register with hazardous content. Adding PubMedQA improved adversarial accuracy for BioBERT and RoBERTa (+14 points) but hurt BERT and DistilBERT, which began missing rephrased hazards, a caution against assuming domain-relevant safe data is always beneficial.

Next steps. If I had more time I would attempt to search for more data to train on. There are a ton of other datasets out there, especially benign ones like BioASQ and MedQA. I would also like to contact the authors of the "Virology Capabilities Test (VCT): A Multimodal Virology Q&A Benchmark" to get their dataset of virology to train against. For the past 4 hours I think I focused too much on increasing test-set accuracy and getting larger models to work, but in the future I would want to focus more on improving the evaluation set. The current results suggest that standard held-out accuracy is not the limiting factor in improving the models. Rather the harder problem is operationally harmful biology requests from benign educational biology. All models struggle with questions like "What is the difference between a virus and bacteria?".